

Final

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Github Repository

Set Up

```
library(tidyverse)
library(here)
library(janitor)

nests <- read.csv(here("data", "nest_data_final.csv"))
```

Problems

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, they used a binomial “log odds” regression to relate and predict (*Recurvirostra americana*) habitat occupation to distance from water edge.

In part 2, they used a Chi-Squared test to compare expected vs observed counts for habitat type and occupation by species.

b. Suggestions for Rewriting

Utilizing binomial logistic regression analysis, we found a significant relationship between distance from the water’s edge and American Avocet (*Recurvirostra americana*) microhabitat use.

Utilizing a Chi-Squared test, we found a significant relationship between bird species (American Avocet, Forster’s Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further Analysis

One additional variable that could influence the probability of American Avocet (*Recurvirostra americana*) microhabitat use is **salinity**. This variable is continuous, numerical (ppm) and often calculated after measuring electrical conductivity in micro Siemens per centimeter ($\mu\text{S}/\text{cm}$). Salinity influences a plethora of biological interactions, specifically dehydration, [bodyweight and feeding/preening behavior in American Avocet chicks](#).

Problem 2. Data Analysis

a. Clean & Wrangle

```
nests_clean <- nests |>
  select(case_control, ant_species, height_cm) |>
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  mutate(ant_name = case_match(
    ant_species,
    "AB" ~ "C.sjostedti",
    "RRB" ~ "C.mimosae",
    "BBR" ~ "C.nigriceps"
  )) |>
  mutate(ant_name = as_factor(ant_name),
         ant_name = fct_relevel(ant_name,
                                "C.sjostedti",
                                "C.mimosae",
                                "C.nigriceps"))

str(nests_clean)
```

```
'data.frame': 270 obs. of 4 variables:
 $ case_control: int 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species : chr "RRB" "RRB" "RRB" "RRB" ...
 $ height_cm : num 392 310 513 154 324 ...
 $ ant_name : Factor w/ 3 levels "C.sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ...
```

```
slice_sample(nests_clean, n = 10)
```

	case_control	ant_species	height_cm	ant_name
1	0	RRB	223.5	C.mimosae

2	0	BBR	141.0	C.nigriceps
3	0	AB	132.5	C.sjostedti
4	0	RRB	404.5	C.mimosae
5	0	BBR	71.0	C.nigriceps
6	0	RRB	132.0	C.mimosae
7	0	RRB	200.5	C.mimosae
8	1	RRB	152.5	C.mimosae
9	0	RRB	117.5	C.mimosae
10	0	RRB	132.0	C.mimosae

b. Response Variable

The 1s and 0s present within the “case_control” column of the data represent if a bird nest was observed (1) or not (0) within a specific tree.

c. Purpose of Study

Tree height is a continuous, numerical variable measured in centimeters. As tree height increases, the probability of nest occurrence would increase because the tree contains more space for nesting and is further away from the ground and therefore potential predators. (Section 3: Results, ¶2)

Acacia ant species is a categorical, nominal variable. The four studied ant species differ in aggression of protecting their host trees, as aggressive protection increases, bird nesting also increases because of this inherent nest protection. (Section 3: Results, ¶3)

d. Table of Models

Model number	Model Description
1	Ants and Height: No Interaction
2	Ants and Height: With Interaction

Model number	Predictor List	Interaction Term?
1	Ant Species, Tree Height	No Interaction
2	Ant Species, Tree Height	With Interaction

e. Running the Models

```
model1 <- lm(case_control ~ ant_name + height_cm,  
             data = nests_clean)  
model1
```

Call:

```
lm(formula = case_control ~ ant_name + height_cm, data = nests_clean)
```

Coefficients:

(Intercept)	ant_nameC.mimosae	ant_nameC.nigriceps
-0.26459	0.16279	0.34048
height_cm		
0.00143		

```
model2 <- lm(case_control ~ ant_name*height_cm,  
             data = nests_clean)  
model2
```

Call:

```
lm(formula = case_control ~ ant_name * height_cm, data = nests_clean)
```

Coefficients:

(Intercept)	ant_nameC.mimosae
0.0046090	-0.0600594
ant_nameC.nigriceps	height_cm
-0.4869777	0.0002531
ant_nameC.mimosae:height_cm	ant_nameC.nigriceps:height_cm
0.0009681	0.0045717

f. Selecting the Best Model

```
AIC(model1, model2)
```

	df	AIC
model1	5	263.7512
model2	7	248.1492

- g. Checking Diagnostics**
- h. Visualizing Model Predictions**
- i. Writing a Caption for Figure**
- j. Calculating Model Predictions**
- k. Interpreting Results**

Problem 3. Affective & Exploratory Visualizations

- a. Comparing Visualizations**
- b. Designing an Analysis**
- c. Checking Assumptions & Running Analysis**
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- e. Writing Caption**
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